

Aakash Choudhary

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RESEARCH INTERESTS

My broader area of research is to work on the existence of primitive pair, the existence of r -primitive k -normal pair and the existence of a finite arithmetic progression of r -primitive elements with first term being k -normal, with prescribed traces and norms over finite fields. Further, numerical calculations are carried in SageMath to provide some illustrations.

RESEARCH THESIS

- **Title of the Thesis:** Existence of Primitive Normal Elements with some Properties in Finite Fields
- **Supervisor:** Prof. R.K. Sharma
- **Institute:** Indian Institute of Technology (IIT) Delhi, New Delhi, India

WORK HISTORY

- **Research Scholar (January 2021 - March 2024)** : Senior Research Scholar, Department of Mathematics, IIT Delhi.
- **Research Scholar (January 2019 - December 2020)** : Junior Research Scholar, Department of Mathematics, IIT Delhi.
- **Assistant Professor (August 2018 -November 2018):** Department of Mathematics, Krishna Institute of Engineering and Technology.

EDUCATION

- **Indian Institute of Technology (IIT) Guwahati** Assam, India
Master of Science, Mathematics and Computing. July 2016 - June 2018
- **Deshbandhu College, University of Delhi** New Delhi, India
Bachelor of Science (Honours), Mathematics. July 2013 - May 2016
- **S.B.N Public School, CBSE Board** Ghaziabad, India
Senior Secondary 2013
- **Aum Sun Public School, CBSE Board** Ghaziabad, India
Secondary School Examination 2011

PUBLICATIONS AND SUBMISSIONS

- **Aakash Choudhary and R.K. Sharma:** Existence of primitive pairs with two prescribed traces over finite fields, *Journal of Algebra and Applications* (2023). doi.org/10.1142/S0219498824502451
- **Aakash Choudhary and R.K. Sharma:** On pairs of r -primitive and k -normal elements with prescribed traces over finite fields. (Communicated)
- **Aakash Choudhary and R.K. Sharma:** Fields having arithmetic progression of r -primitive and k -normal elements with prescribed traces. (Communicated)
- **Aakash Choudhary and R.K. Sharma:** Fields having primitive pairs with prescribed norms. (Communicated)

TEACHING EXPERIENCE

● Teaching Assistant

- **Representations of Finite Groups:** Fall 2020, IIT Delhi.
- **Calculus:** Fall 2019 & Fall 2023, IIT Delhi.
- **Cryptography:** Spring 2020, IIT Delhi.
- **Linear Algebra and Differential Equations:** Spring 2021 - 2022, IIT Delhi.
- **Abstract Algebra:** Spring 2022, IIT Delhi.
- **Linear Algebra:** Fall 2021, IIT Delhi.

CONFERENCES AND WORKSHOPS

- **Journées Arithmétiques 2023:** Presented my paper entitled “*Existence of primitive pairs with two prescribed traces over finite fields*” at the Conference *Journées Arithmétiques 2023* held at the University of Lorraine, Campus Aiguillettes, Nancy, France from 3 July 2023 to 7 July 2023.
- **38rd Annual Conference of the Ramanujan Mathematical Society:** I have presented a paper titled “Fields having primitive pairs with prescribed norms” at the Department of Mathematics, IIT Guwahati, during 22nd December - 24th December, 2023.
- **International Meet on Number Theory:** Participated in the International Meet on Number Theory to commemorate the 135th birth anniversary of Srinivas Ramanujan held at IIT Kanpur during 27th February 2023 to 1st March 2023.
- **Analytic number theory:** Attended a workshop on ‘*Analytic number theory*’ organized by Kerala School of Mathematics (KSOM) from 15th Jan to 24th Jan 2023.

PROJECTS

- **Hardy Hilbert Spaces & Toeplitz Operators :** I have made a Project titled as above, enlightening Elementary introduction to Hardy-Hilbert Space and Spectral Properties of Toeplitz Operators under guidance of Dr. Jitendriya Swain Sir, Assistant Professor, Department of Mathematics, IIT Guwahati.
- **Abelian Forcing Set:** I have given a Seminar on Abelian Forcing Set. *i.e* if $(ab)^n = a^n b^n$ for all a, b belonging to a group G then which property of these n 's would imply that G is abelian at IIT Guwahati.

SCHOLASTIC ACHIEVEMENTS

- **Research Scholar Travel Award:** Awarded Research Scholar Travel Award (RSTA) of INR 1.5 lacs, IIT Delhi for presenting my research work at the Conference *Journées Arithmétiques 2023* Nancy, France.
- **Senior Research Fellowship:** Awarded Senior Research Fellowship for Ph.D. in 2021.
- **Junior Research Fellowship:** Awarded Junior Research Fellowship for Ph.D. in 2019.
- **National Eligibility Test:** Qualified National Eligibility Test, Mathematical Science (NET-CSIR) for Lecturership in June 2018.
- **Graduate Aptitude Test in Engineering:** Qualified Graduate Aptitude Test in Engineering, Mathematical Science (GATE-MS) in February 2018.
- **Junior Research Fellowship:** Qualified Junior Research Fellowship, Mathematical Science (JRF) in December 2017.
- **Joint admission test for Masters:** Qualified Joint admission test for Masters (IIT-JAM) Mathematics in 2015.

EXTRA CURRICULARS

- Volunteered SPIRIT 2017 (The Annual Sports Fest of IIT Guwahati).

TECHNICAL EXPERTISE

- **Languages & Frameworks:** Proficient in Matlab, L^AT_EX, SageMath.

REFERENCES

- Prof. R.K. Sharma
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